

Data organization

File	Purpose
student_management.py	Menu, validation, CRUD operations, custom exceptions, and logging.
students.csv	Core fields: registration number, first name, last name, and email.
students.json	Additional details keyed by registration number: address, contact, and program.
student_system.log	Time-stamped successful actions, warnings, startup/shutdown events, and errors.

The `main()` function maps menu choices to focused handler functions. This keeps the loop readable and makes the storage class reusable in automated tests or a future graphical interface.

4. Testing Results

Test	Expected result	Outcome
Load sample data	Four linked records display	Pass
Add and search	New record stored and retrieved	Pass
Update email	CSV value changes; JSON remains linked	Pass
Duplicate registration	DuplicateStudentError raised	Pass
Delete and re-search	Both stores remove record; not-found raised	Pass
Invalid menu choice	Friendly warning; program continues	Pass

The completed system meets the assignment requirements while separating user interaction, validation, storage, error handling, and logging. The files are portable because all runtime paths are resolved relative to `student_management.py`.